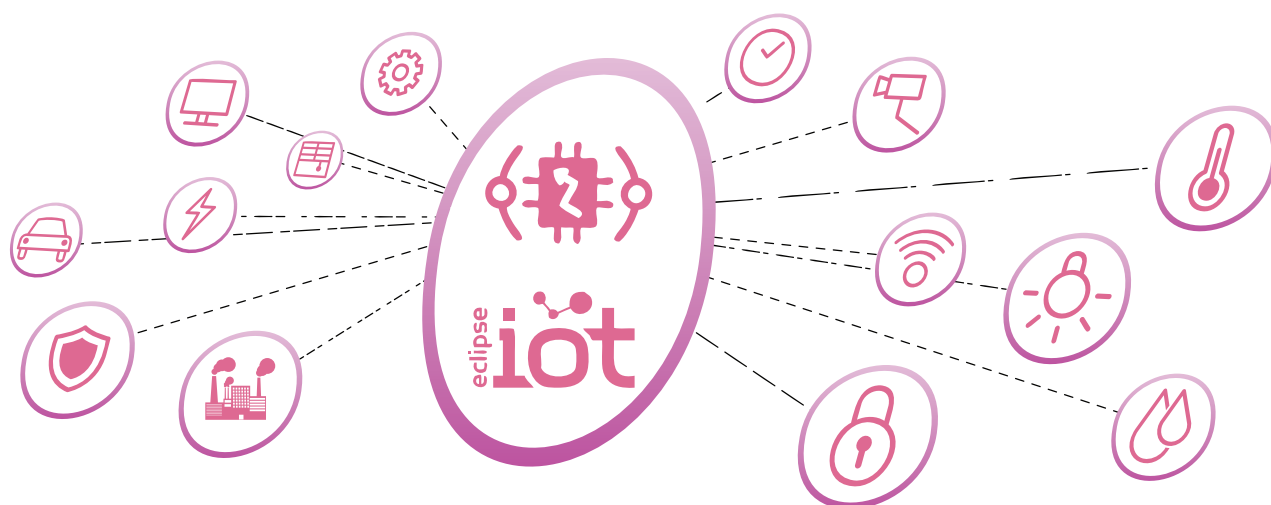


How Open Source is Helping IoT Evolve



Let's delve into the essence of IoT, explore its profound impact across diverse industries, and introduce the pivotal role of open source software in shaping its evolution.

IoT, short for the Internet of Things, encompasses a vast network of interconnected physical entities objects, devices, vehicles, and so on. These entities are equipped with sensors, software, and network connectivity, allowing them to autonomously collect, exchange, and analyse data. This enables them to interact with and respond to their environment. The fusion of the physical and digital realms redefines the way we gather, process, and harness information, ushering in a new era of connectivity and intelligence.

IoT's significance in industries and open source software's impact

IoT's significance transcends mere

technological innovation, promising to revolutionise a multitude of industries such as healthcare, agriculture, manufacturing, and transportation. In healthcare, IoT devices offer real-time patient monitoring, elevating the quality of care and enabling remote patient oversight. Agriculture benefits from IoT-powered sensors optimising crop management and sustainable irrigation practices, while manufacturing gets efficient through predictive maintenance and downtime reduction. These examples underscore IoT's impact on precision, efficiency, cost-effectiveness, safety, and sustainability across sectors.

Meanwhile, open source software lies at the core of IoT's rapid evolution. It adopts a collaborative ethos, providing freely accessible source

code for software modification and distribution. Open source communities have been instrumental in driving IoT innovation, sharing knowledge, and creating versatile tools and platforms.

The choice of technologies and tools can significantly impact the success of a project. Open source technologies and tools have gained prominence for their flexibility, affordability, and vibrant communities.

Overview of key open source IoT platforms

Open source IoT platforms are the backbone of many IoT applications, providing the infrastructure for data collection, processing, and device management. Here are some popular open source IoT platforms.